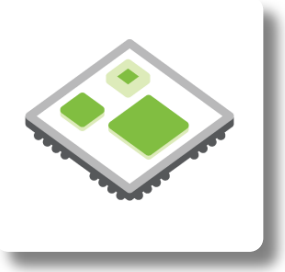
SERVER:
OPEN CHIPLET
ECONOMY



Bunch of Wires Flexi 1.0 : a low cost- low power D2D interface

Morgan Whately

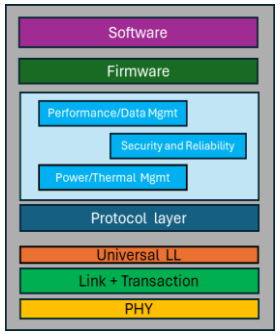
Distinguished Engineer, Infineon Technologies

Kash Johal

Founder, Yorchip

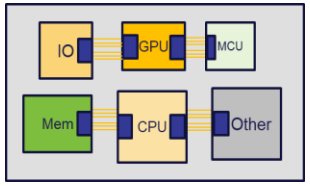


OCP Open Chiplet Economy (OCE)

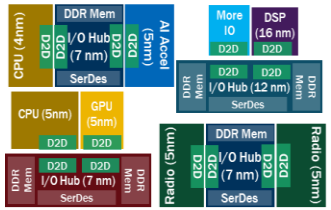


System Stacks

AI & HPC Modularity

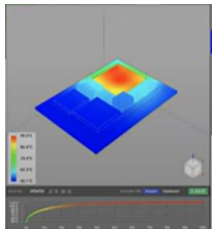


Foundation Chiplet Systems Architecture



Algorithm Decomposition



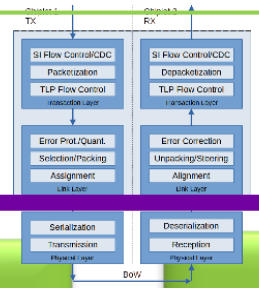


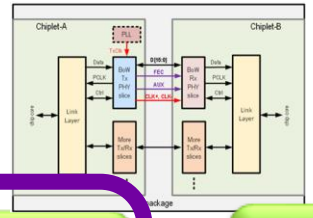
Automation for Chiplet Design

Virtual Prototyping

Design Tools

Omni Connect





Logical Interconnects

BoW Phy

Optical Interconnects

Interconnects (D2D & Optical)

BoW.Flexi Introduction

- BoW.Flexi is a low overhead chiplet interface developed for communication where simplicity, efficiency, and cost are optimized over absolute bandwidth.
- 8 Commands supported
  - Configuration command is used for setup (No sideband required)
- Minimum 6 pin interface
  - Single Data Rate – any speed clock
  - Frame signal used to support variable length packets
  - Electricals based on the BoW Sideband interface – 0.65 to 0.95V operation
- Simplest implementation is a memory mapped interface
- Optional Fields can be enabled to support protocols e.g. AMBA® AHB/APB, AXI

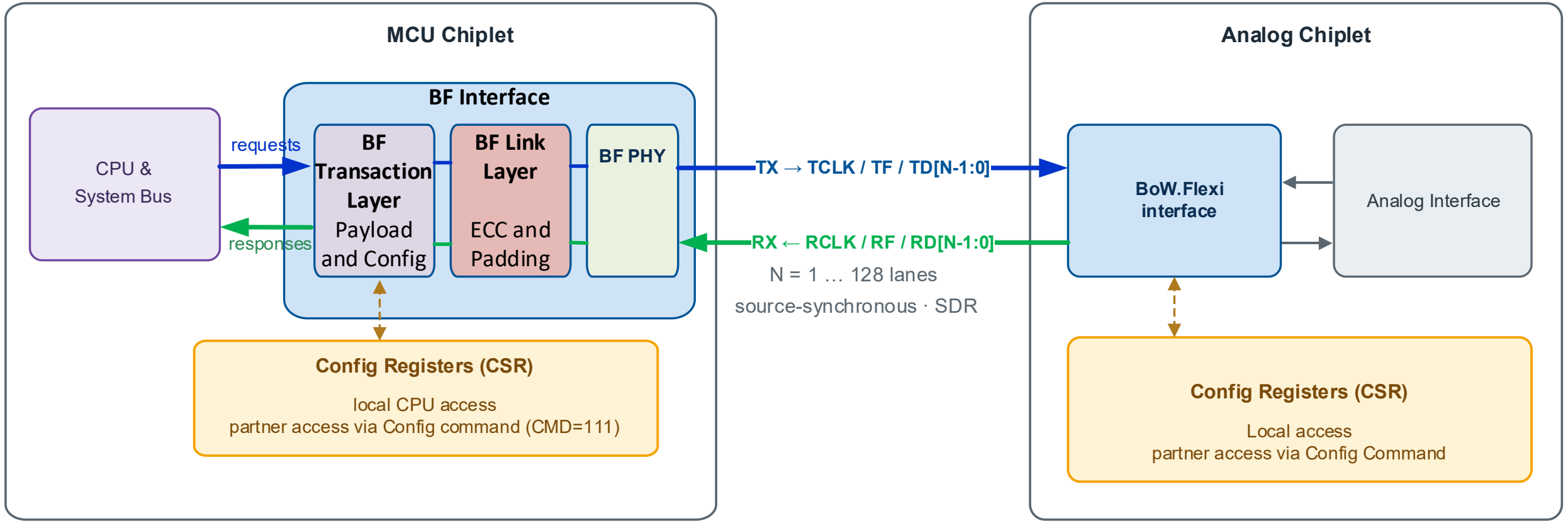
BoW.Flexi: Comparison to BoW 2.0

Key Specs	# Lanes	Link Training	Sideband Required	Frequency In Gbs/pin	ESD	System Power Modes	Packaging
BoW 2.0	8,16 DDR	Yes	Yes	2/4/8/16/32	50V	Active	Advanced/ Standard
BoW.Flexi	1 – 128 SDR	No	No	0 to 4 Implementation dependent	Any V Flexible	Active Standby Deep-Sleep	Wire bond Flip Chip Hybrid etc

- BoW.Flexi is optimized for low power, low latency
- BoW.Flexi is a simpler PHY with SDR clocking and no serialization
- Bow.Flexi is simple to implement across multiple nodes and for many markets
- BoW.Flexi is optimized for interoperability with plans for reference RTL

BoW.Flexi: a simple, configurable chiplet link

BoW.Flexi — a simple, configurable chiplet-to-chiplet link
Both sides of Link can Initiate transactions



Simple Packet Structure

Every packet starts with a 3-bit CMD. The payload that follows depends on the command. Only enabled fields are present; PAD aligns the packet to the lane width.

CMD	Command	Class	Payload (after CMD)
000	Non-Posted Write	Sys	[OPT0–3] SIZE ADDR DATA [OPT_T]
001	Read	Sys	[OPT0–3] SIZE ADDR [OPT_T]
010	Read Response	Sys	[OPT0–3] DATA [OPT_T]
011	Write Response	Sys	[OPT0–3] [OPT_T]
100	Posted Write	Sys	[OPT0–3] SIZE ADDR DATA [OPT_T]
101	Virtual Wire	Link	VW state (1–32)
110	Extension	Sys/Adv	EXTCMD [OPT0–3] SIZE DATA [OPT_T]
111	Config / Error	Link	LEN WR RD_RESP ADDR DATA

[OPT0–3] = the four leading optional fields (after CMD); [OPT_T] = the single trailing optional field (end of payload). Each is enabled per command via opt_fieldN_cmd_support / opt_field_t_cmd_support. EXTCMD = 4-bit command-extension on CMD=110. ADDR width 1–64 b; DATA up to 1024 b.

Common system packet shape



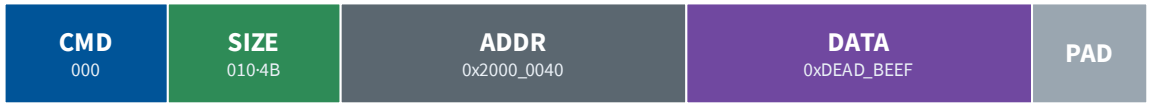
† SIZE is not on every command. Responses (010/011) carry no SIZE, and fixed_size_enable can omit it from all system commands — the receiver then infers data length from the fixed_size register.

Two Link commands use a different shape: Virtual Wire (CMD + wire state) and Config/Error (CMD + LEN + WR + RD_RESP + ADDR + DATA).

Command examples - system interface

Example link config: 32-bit address, 32-bit data, SIZE=010 → 4 bytes, no optional fields. Only enabled fields appear; PAD aligns to the lane width.

CMD 000 Non-Posted Write — SIZE + ADDR + DATA — bridge holds HREADY low until Write Response (011).



CMD 001 Read — SIZE + ADDR only — no DATA; target replies with a Read Response (010).



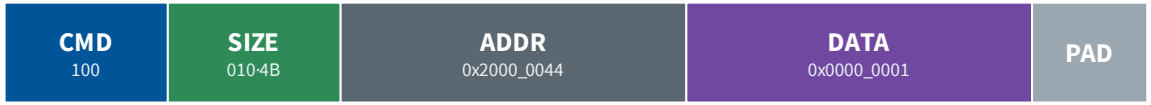
CMD 010 Read Response — DATA only — supplies HRDATA back to the requesting master.



CMD 011 Write Response — CMD only — arrival implies HRESP=OKAY and completes a non-posted write.



CMD 100 Posted Write — Same shape as 000, but the master does not wait — response is flow-control.



Command examples –Config & Virtual Wires

CMD 111 Config / Error

Reads/writes the partner's registers; the same packet reports errors (WR=1 to the Error register).



LEN selects DATA width (00=8, 01=16, 10=32 bit), WR picks read vs. write, RD_RESP marks a Config Read Response. Config writes carry no acknowledgement.

CMD 101 Virtual Wire

CMD + the virtual-wire state (1–32 bits). VW bits are real data and are never padded inside the vector.



With x8 lanes, CMD + the first 5 wires fill beat 0 (no padding allowed inside the wire vector).

Both are Link Interface commands: Virtual Wire and Config/Error are managed by the Transaction Layer, which enforces the single-outstanding policy for the Link Interface.

Publication of Bow.Flexi Version 1.0

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OPEN Compute Project

BoW.Flexi Project: The Open Chiplet Economy BoW PHY Workstream

BoW.Flexi

Version 1.0.0

Modular Base Specification

Effective 17th September 2026

Author: Morgan Whately, Infineon Technologies

https://www.opencompute.org/documents/bow-flexi-1p0-final-pdf

BoW PHY Call to Action

- Join the BoW working group to contribute to the BoW D2D specifications!
  - BoW is fully openly licensed and available to all for adoption and participation
  - BoW Workstream Group meets every Wednesday at 9am Pacific

  Zoom link: https://opencompute-org.zoom.us/j/87373664893?pwd=UURoVlBUTWJHWWFsWUZNeVJKdlpadz09

  OCE: https://www.opencompute.org/community/open-chiplet-economy (calendar at bottom of page)

- Where to find additional information (URL links):
  - BoW.Flexi SPEC, SVR, BoW Flexi V1.0.0:
  - https://www.opencompute.org/documents/bow-flexi-1p0-final-pdf
  - BoW 2.0 PHY Spec: https://www.opencompute.org/chiplets/26/bunch-of-wires-bow-phy-specification-20

Thank You!

